



IDEA LAB

(Soil Moisture Detector)

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Introduction:

Soil moisture detection plays a crucial role in agriculture, gardening, and civil engineering, as it provides valuable information about the water content in the soil, which directly affects plant health, irrigation planning, and construction projects. Proper moisture levels are essential for optimal plant growth, and detecting these levels can help avoid overwatering or underwatering.

This project focuses on the design and implementation of a soil moisture detection device that determines the moisture content in the soil and provides feedback through sound and light. The system detects three levels of moisture—high, medium, and low—and indicates these levels by varying the intensity of an LED light and the volume of a buzzer. The intensity of the light and the volume of the sound are directly proportional to the moisture level in the soil, offering an easy and effective way to monitor soil conditions. This system is valuable for ensuring efficient water usage and supporting sustainable agricultural practices.

Objective:

To design and implement a soil moisture detection system that:

- Remains inactive when soil is dry.
- Provides moderate feedback (dim LED and mild buzzer sound) for medium moisture.
- Activates fully (bright LED and loud buzzer) for high soil moisture.

Project: Soil Moisture Detector with LED and Buzzer

Aim: To design and implement a soil moisture detection system using a LED and a Buzzer.

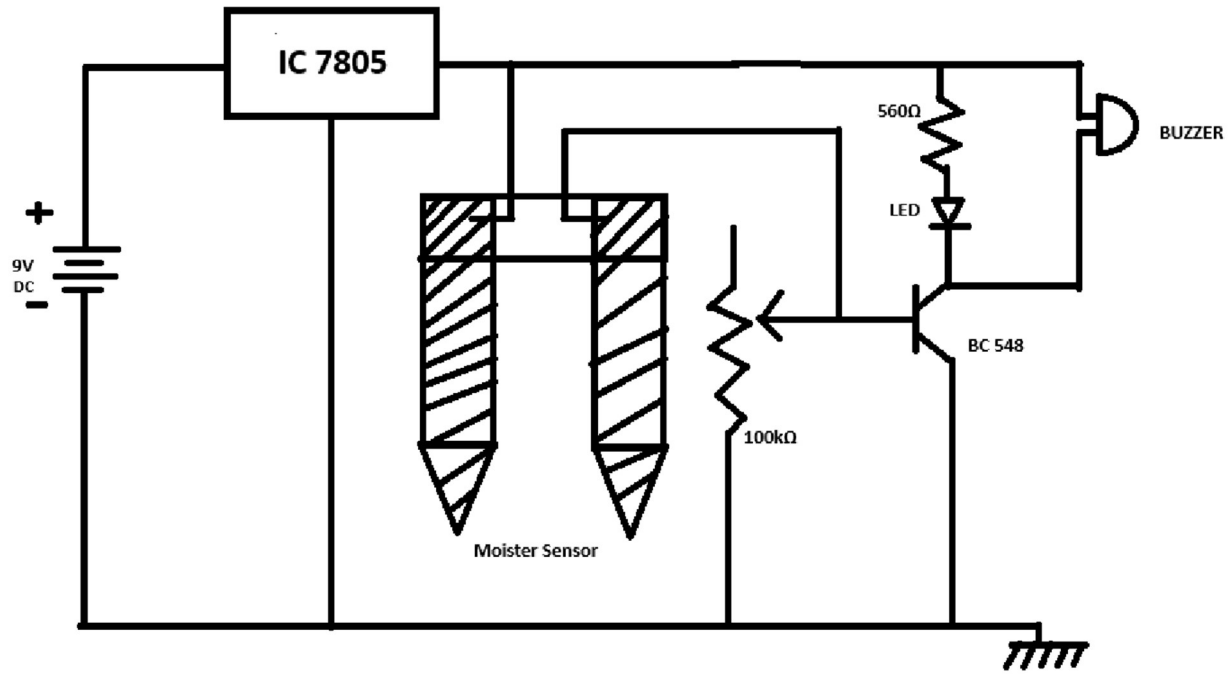
Items Required:

- IC7805 (Voltage Regulator)
- BC548 (NPN Transistor)
- Moisture Sensor
- LED (Light Emitting Diode)
- Buzzer
- Resistors (560Ω and $100k\Omega$)
- 9V Battery
- Copper PCB
- Photo Paper (for heat transfer)
- FeCl_3 (Ferric Chloride, for etching the PCB)
- Electric Iron (for heat transfer to PCB)
- PCB Drill (for creating holes in the PCB)
- Soldering Iron (for soldering the components)
- Wires and Connectors (for wiring the components together)

Operation:

1. Design the circuit in AutoDesk Eagle
2. Print the circuit
3. Install the components
4. Test the circuit

Circuit Diagram:



Working Principle:

The circuit works based on the principles of conductivity and transistor switching. Here's a breakdown:

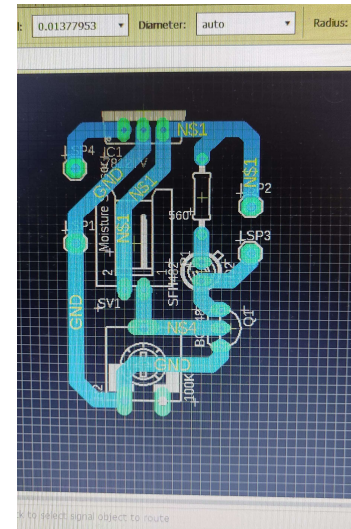
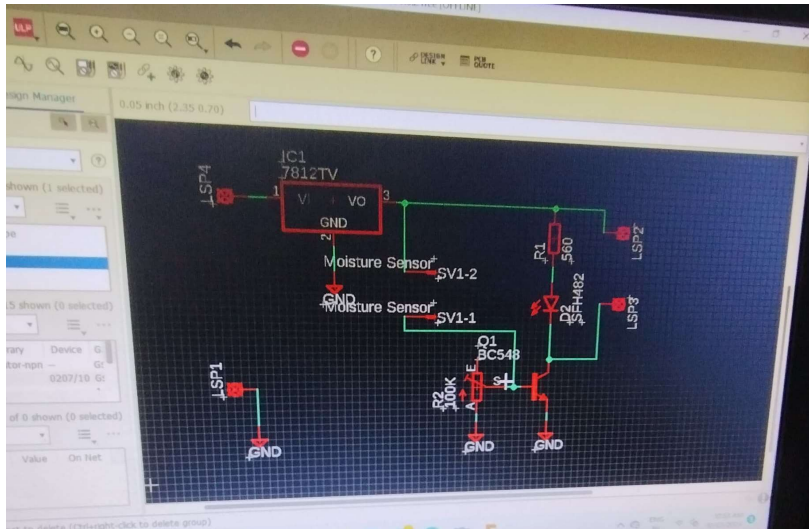
1. **IC7805 (Voltage Regulator):**
 - a. Regulates the input voltage (9V) to a stable 5V output, which powers the moisture sensor and the circuit.
 - b. Ensures consistent operation despite fluctuations in the input supply.
2. **Moisture Sensor:**
 - a. Detects soil moisture levels and converts them into voltage signals.
 - b. Higher moisture increases conductivity, allowing more current to flow.
3. **BC548 Transistor (Switching Mechanism):**
 - a. Acts as a switch to control current flow through the LED and buzzer based on the moisture sensor's signal.
 - b. Operates in three states depending on the base voltage:
 - i. **Low Moisture:**
 1. Base = 0V; no current flows from the base to the emitter.
 2. The transistor remains OFF, so no current flows from the collector to the emitter.

3. LED and buzzer remain inactive.
- ii. **Medium Moisture:**
 1. Transistor partially conducts.
 2. The transistor partially turns ON, allowing a controlled amount of current to flow from the collector to the emitter.
 3. The LED glows dimly, and the buzzer produces a mild sound.
- iii. **High Moisture:**
 1. Transistor fully switches ON.
 2. Maximum current flows, brightening the LED and producing a loud buzzer sound.
4. **Variable Resistance (100k Ω):**
 - a. Adjusts the sensitivity of the base voltage.
 - b. Ensures the circuit responds accurately to varying moisture levels, avoiding false triggering.
5. **Current Flow and Output Response:**
 - a. The moisture sensor output determines the base current of the BC548.
 - b. As the soil moisture increases, the base current increases, enhancing the collector-emitter conduction.
 - c. This gradually activates the LED and buzzer, providing proportional feedback for low, medium, and high soil moisture.

Procedure:

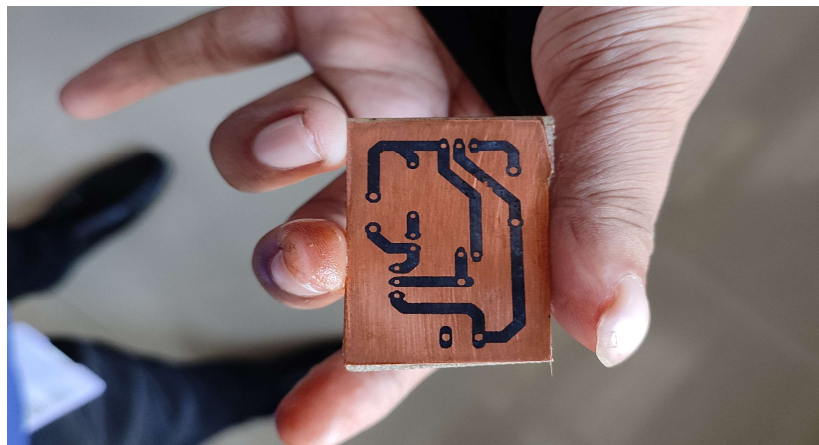
1. Making Schematic and Board in Autodesk Eagle:

1. Open Autodesk Eagle and create a new project.
2. Enable the grid system to ensure accurate placement of components.
3. Design the circuit schematic by adding components (IC7805, BC548, moisture sensor, LED, buzzer, resistors, and 9V battery) and connecting them using the "Net" tool.
4. Perform a **Design Rule Check (DRC)** to identify and fix any errors in the schematic, ensuring all connections are correct.
5. Switch to the PCB layout editor and arrange the components compactly while maintaining proper trace spacing to prevent short circuits.
6. Use the grid to align components neatly and make the layout user-friendly.
7. Export the PCB layout and print it on photo paper using a laser printer.



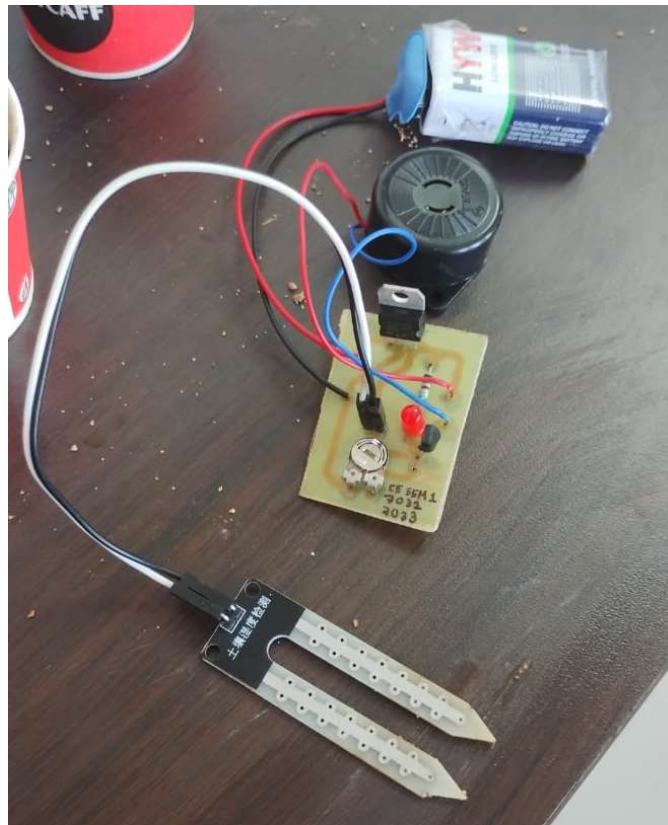
2. Fabricating the PCB:

8. Attach the printed photo paper to a copper-clad PCB using heat transfer with an electric iron until the paper turns black, indicating proper adhesion.
9. Soak the PCB in water to remove the photo paper, leaving the inked circuit layout on the copper.
10. Dip the PCB in Ferric Chloride (FeCl_3) solution and jerk it periodically for 25–45 minutes to dissolve unwanted copper, leaving only the circuit pattern.
11. Rinse and clean the PCB thoroughly.
12. Then drill holes at component points using a PCB drill.



3. Assembling the Circuit:

13. Insert the components (resistors, IC7805, BC548, moisture sensor, LED, buzzer) into their respective positions on the PCB.
14. Solder the components securely, ensuring proper connections without short circuits.
15. Connect the 9V battery and test the circuit for functionality.



Circuit Analysis:

TABLE1: Reading of IC 7805

Parameter	Reading	Theoretical
INGOING VOLTAGE	8.05V	9V
GROUND	0V	0V
OUTGOING VOLTAGE	5.01V	5V

TABLE2: Reading of Transistor BC578

Parameter	High	Medium	Low
Collector	0.03V	2.86V	4.57V
Base	0.747V	0.646V	0V
Emitter	0V	0V	0V

1. Power Supply (IC7805 and 9V Battery):

- A 9V battery serves as the power source.
- The IC7805 voltage regulator ensures the circuit operates at a steady 5V output, which is critical for consistent performance of the components.
- Input voltage to the IC is 8.50V, and the output is regulated to 5.01V.

2. Moisture Sensor:

- The moisture sensor consists of two conductive probes that are inserted into the soil.
- When soil moisture is high, it conducts electricity better, creating a lower resistance path between the probes. In dry soil, the resistance is high, and current flow is minimal.

3. Variable Resistor (100kΩ):

- The variable resistor (potentiometer) is used to calibrate the sensitivity of the circuit to different moisture levels. It controls the base current of the BC548 transistor.

4. BC548 Transistor (NPN):

- **Dry Soil (Low Moisture):**
 - Minimal current flows through the moisture sensor. This results in negligible base current for the transistor. As a result, the transistor remains in the *off state*, and the LED and buzzer are inactive.
- **Medium Moisture:**
 - Moderate current flows through the sensor, providing a partial base current to the transistor. The transistor enters the *active region*, allowing some current to

flow from the collector to the emitter. This causes the LED to glow dimly and the buzzer to beep moderately.

- **High Moisture:**

- The sensor conducts maximum current due to the reduced soil resistance. This drives sufficient base current into the transistor, saturating it. In the *saturation region*, the transistor allows full current flow, causing the LED to glow brightly and the buzzer to sound at full volume.

5. **LED and 560Ω Resistor:**

- The LED provides a visual indication of soil moisture levels. The 560Ω resistor limits the current through the LED to prevent damage.

6. **Buzzer:**

- The buzzer provides an auditory indication of moisture levels. It is connected to the same collector-emitter circuit as the LED, sharing the transistor's current output.

7. **Current Flow through the Circuit:**

- The moisture sensor generates a voltage signal proportional to soil conductivity. This signal is amplified by the BC548 transistor, which controls the LED and buzzer based on the moisture levels.



For visual demonstration [Scan The QR Code](#)



Error Analysis:

During initial testing, the circuit failed to function correctly. Upon analysis, it was discovered that the IC7805 had been short-circuited during soldering. The short circuit was fixed by desoldering and resoldering the IC properly. After this correction, the circuit worked as expected, successfully detecting soil moisture levels and providing the desired outputs.

Precautions:

- Handle the soldering iron and FeCl₃ solution carefully to avoid burns or injuries.
- Ensure components like the transistor and moisture sensor are correctly oriented.
- Avoid overheating components during soldering to prevent damage.
- Test the circuit on a breadboard before final assembly.

Conclusion:

The soil moisture detection system was successfully designed and implemented to monitor the moisture content in the soil. By utilizing a moisture sensor, BC548 transistor, IC7805 voltage regulator, and other components, the circuit effectively demonstrated the following:

- **Real-Time Moisture Monitoring:**
The system provided distinct feedback for three moisture levels (dry, medium, and high) using an LED and a buzzer. This makes it practical for applications like gardening, agriculture, or soil health monitoring.
- **Efficient Component Integration:**
The use of a variable resistor allowed precise calibration of the circuit, ensuring accurate response to varying soil conditions. The BC548 transistor acted as a reliable switching and amplification component.
- **Scalability and Practicality:**
The circuit design and analysis highlight its potential for integration into automated irrigation systems, enhancing water efficiency by signaling the need for watering based on soil conditions.
- **Applications as a Civil Engineer:**
The system is particularly beneficial for civil engineers working on projects such as irrigation planning, soil stabilization, and landscaping. By providing real-time data on soil moisture, the system aids in designing sustainable water management solutions and ensures the structural integrity of soil in construction projects.

The system's performance after fixing initial soldering errors validated the circuit's robustness and sensitivity to soil moisture levels. Overall, the project demonstrated a clear understanding of electronic components, circuit design, and practical implementation, providing a foundation for further advancements in smart agricultural technologies.